

Why a routing graph

Global routers do not hop pixel by pixel

The idea

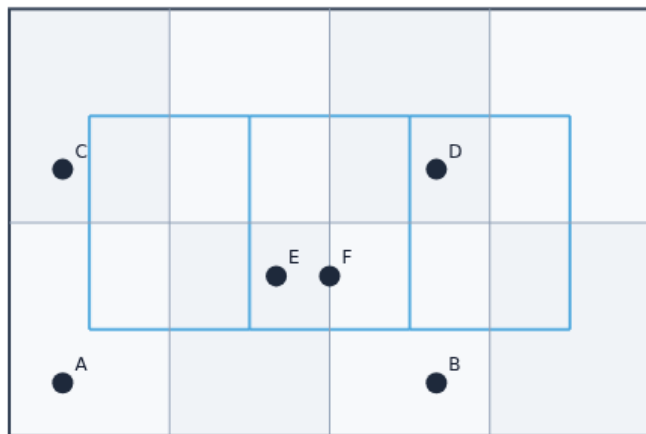
- To the upper neighbor when j plus one is less than n_y
- On four by two you get eleven edges
- Store each edge as a sorted pair of GCell indices so direction does not matter

GCell graph

ROUTING GRAPH

GCell graph

Step 1 / 5 · grid · module01-01-routing-graph



Explanation

Four-by-two GCells form a grid graph: nodes are tiles, edges are adjacency.

- 10 edges total
- Cap=2 per edge

edges: 10

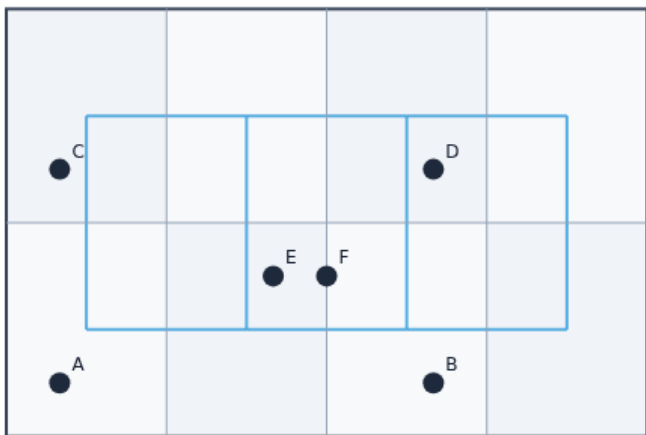
GCell graph

Horizontal edges

ROUTING GRAPH

Horizontal edges

Step 2 / 5 · h · module01-01-routing-graph



Explanation

Three horizontal edges per row connect columns i and $i+1$.

- 6 horizontal
- Same j

Row bands

Horizontal edges

Vertical edges

ROUTING GRAPH

Vertical edges

Step 3 / 5 · v · module01-01-routing-graph



Explanation

Four vertical edges connect rows j and $j+1$.

- 4 vertical
- Same i

Column risers

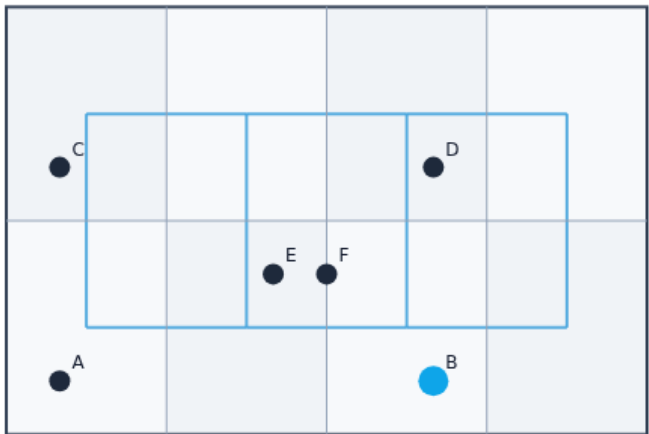
Vertical edges

Neighbors

ROUTING GRAPH

Neighbors

Step 4 / 5 · neigh · module01-01-routing-graph



Explanation

Interior GCell (1,0) has three neighbors; corners have two.

- Degree depends on location
- B sits near mid-bottom

mid neighbors=3

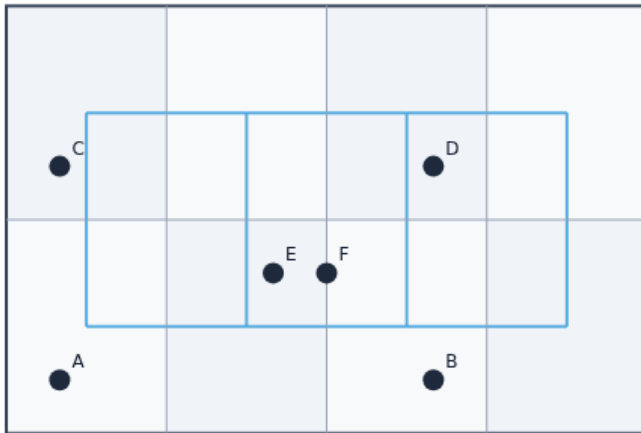
Neighbors

Capacity

ROUTING GRAPH

Capacity

Step 5 / 5 · cap · module01-01-routing-graph



Explanation

Each edge tracks usage vs capacity—overflow drives rip-up.

- Cap=2
- Document goldens

Router contract

Capacity

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / GCell routing graph

INTERACTIVE TOOL

GCell routing graph

Four-by-two GCell grid with horizontal and vertical edges; capacity two per edge.

Your job: explore the routing graph. Edge count is 10. Interior GCells have up to four neighbors; corners have two.

Reset workspace

Challenges0 / 8 cleared

Pick one[Intro] Ten edges

Pick a challenge and click **Start**. Move cells, route, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A

B

C

D

E

F

←

→

↑

↓

Reset to starter

Reveal golden (study)

Route L-HV

Route L-VH

Route maze

Rip-up reroute

Graph

Reference / metrics

selected: A @ GCell (0,0)
routes: (none — click Route)
overflow total: 0.00 · max: 0.00 · count: 0
edge capacity: 2
HPWL: 34.0
view: your placement

Browser lab starter

learn_global_routing — 01 routing graph

Implement track

- Implement `edge_list(nx, ny)` in `common/grutil.py`
- Print all edges for the tiny instance
- Verify $((0,0),(1,0))$ is present and diagonals are absent

Pitfalls

- Counting tile interiors instead of adjacency channels
- Double-counting undirected edges as two directed arcs without collapsing
- Forgetting top and right boundary checks

Your turn

- Finish the checklist
- Sketch the eleven edges from memory
- Next: map pin placements to terminal GCells